SDC HACKATHON 2026 · PHASE 1 – PRELIMINARY PROPOSAL (PRODUCT REQUIREMENT DOCUMENTATION)

Future Edge

Your career weather forecast for the AI era · Team BEBZA · Domain 1 – Aiversecity Future-Ready Workforce

1. Team Overview and Project

Team Name	BEBZA
Selected Domain	Domain 1 – The Aiversecity Future-Ready Workforce
Selected SDG Track (1–17)	SDG 8 – Decent Work and Economic Growth (targets 8.2, 8.5), supported by SDG 4 – Quality Education (targets 4.3, 4.4)
Project / Solution Name	Future Edge
Name (Leader)	Azimkhan Bekassyl
Email & Phone (Leader)	beka2007azimut@gmail.com · +7 701 770 7593
University Affiliation(s)	Xiamen University Malaysia (XMUM)
Number of Team Members	5 (Azimkhan Bekassyl, Elnurken Abdigaparov, Bernar Zholdybek, Alibek Berik, Shirvanskiy Zeinal)
Links	Live prototype: https://sdc-2026-future-edge.replit.app Code: https://github.com/Magisterofdrinkingcoffee/sdc2026-future-edge

2. Executive Summary & Problem Statement

2.1 The Problem Context [max 250 words]

AI is changing which jobs exist faster than training programmes can be updated. In Malaysia, 56% of professionals already use AI tools at work, yet only 26% have received formal training (Hays Asia Salary Guide 2025), and the Ministry of Science, Technology and Innovation expects more than 600,000 workers to need reskilling within three to five years. The World Economic Forum's Future of Jobs Report 2025 (Malaysia profile) projects that the share of work tasks done mainly by people will fall from 47% to 33% by 2030, and that data-entry clerk jobs will shrink by 28% while AI and machine-learning specialist jobs grow by 29%.

The people most affected are not short of information about AI; they are short of a **personal, forward-looking answer**. A fresh graduate choosing a first job, an admin executive whose spreadsheet work is being automated, or a final-year engineering student cannot tell which of *their own* skills AI will take over, which will keep them valuable, and what single thing to learn next. Existing career tools give a one-time keyword match against a static list, and "AI career coaches" are chatbots that can invent jobs or courses.

This matters economically (SDG 8: productive employment) and socially (SDG 4: relevant skills and lifelong learning), most of all for workers without paid career advice. The specific moment we design for is: **a person deciding what to learn next, before AI makes that decision for them.**

2.2 The Proposed AI Solution [max 300 words]

Future Edge turns a CV into a **three-year career forecast**, shown as a weather forecast: an Edge Score from 0 to 100 for each year, meaning "how hard you will be to replace by AI". A user goes from CV to ranked roles and an ordered learning plan in about three minutes.

Three models work together:

- **Model 1 – AI exposure:** a text classifier trained on 19,265 human-rated O*NET work tasks (Eloundou et al., 2023) reads each skill and the user's own CV sentence and estimates how much of that work AI can do (73.4% accuracy on occupations it never saw, versus 53.2% for a simple baseline).
- **Model 2 – AI-trend signal:** monthly arXiv paper counts for ten AI-capability areas, forecast with a damped trend and an 80% range, decide how fast AI tools arrive for each skill. Refreshing the data live changes the forecast.
- **Model 3 – Catch the AI:** instead of a self-rating quiz, the user reviews an AI-written work sample with planted mistakes. This measures judgment and responsibility, the skills that keep humans in charge of AI.

These combine with real Malaysian job-market data (WEF Future of Jobs 2025 Malaysia profile, DOSM wages, TalentCorp MyCOL critical occupations) into an explainable score. Every role and skill is tagged to SDG targets, the user sees moderate and fast AI-disruption scenarios side by side, and the plan starts with the "One Move" that raises their score most.

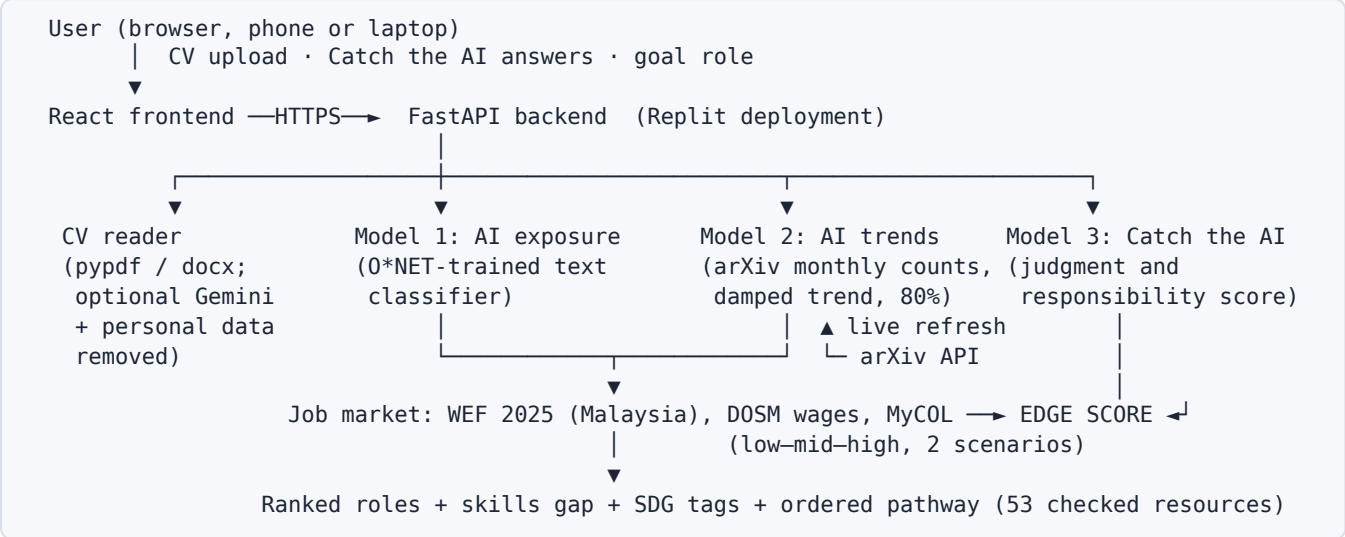
No generative model writes recommendations, so the system cannot invent jobs, courses or statistics. From the SDC Phase 1 workshops we use **Replit** for development and hosting and **structured outputs** (a JSON schema) when the optional Gemini API reads hard-to-parse CVs; every returned phrase is checked against the CV text.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

Layer	Technology
Frontend	React 18, Vite, Tailwind CSS (mobile-friendly, step-by-step wizard, forecast and quest map)
Backend / API	Python 3, FastAPI, Uvicorn – endpoints for CV upload, forecast, pathway, Catch the AI and trend refresh
AI models	Model 1: TF-IDF + logistic regression (trained with scikit-learn on CPU, exported to JSON, runs in plain Python). Model 2: damped log-linear trend with 80% intervals. Model 3: exact scoring of a performance task.
Data store	Versioned JSON catalogs in the repository: 52 skills, 34 roles, 53 learning resources, SDG targets, trend snapshot. No personal data is stored.
External data	arXiv API (trend signal, refreshable), O*NET / GPTs-are-GPTs labels, WEF Future of Jobs 2025, DOSM, TalentCorp MyCOL
GenAI API (optional)	Google Gemini API with a JSON schema (structured outputs) for photo or complex-layout CVs

Cloud hosting	Replit autoscale deployment (build.sh → start.sh)



Did your project utilise any official SDC sponsor-provided GenAI API or Cloud Platform?
Yes – **Replit** (official SDC sponsor). We used Replit to import and run the project, install dependencies, run the test suite and the course-link checker (which needs open internet), and to host the live prototype as a Replit autoscale deployment at https://sdc-2026-future-edge.replit.app. Secrets such as the optional Gemini key are kept in Replit Secrets. RunPod was not used because no part of the project needs a GPU (Participant Handbook 9.4): all models train and run on a CPU in seconds.

3.2 Design-Thinking & Deployment Strategy [max 200 words]

Empathise: we wrote three realistic personas from the problem statement – a final-year engineering student, an admin executive with six years' experience and a fresh CS graduate – and built synthetic CVs for each. **Define:** their shared question was not "which jobs match my keywords?" but "which of my skills will still be mine in three years, and what do I learn first?" **Ideate:** we rejected a chatbot (it can invent jobs and courses) in favour of a forecast built from measurable signals, and replaced self-rating quizzes with a test that has known answers. **Prototype and test:** every model has a held-out evaluation, 49 automated tests guard the scoring rules, and every course link was checked on 24 October 2026. Testing showed that a plain trend over-shot, so we switched to a damped trend; it also showed leadership being flagged "at risk", so human skills now gain AI exposure more slowly.

Deployment readiness: the prototype is already live on Replit. For Phase 2 we will run a user test with XMUM students, add a Bahasa Malaysia interface and an employer team view, and keep the same deployment pipeline.

4. Preliminary Environmental & SDG Impact Framework

Metric Category	Our Response
Target SDG Sub-Target(s)	**Target 4.4** – "By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship." **Target 8.5** – full and productive employment and decent work for all (supporting target).

Current Problem Baseline	Only **26%** of Malaysian professionals who use AI at work have received formal AI training, while 56% already use AI tools (Hays Asia Salary Guide 2025, cited in the Domain 1 brief). More than **600,000** workers need reskilling within three to five years (MOSTI). The share of tasks done mainly by people is expected to fall from **47% to 33%** by 2030 (WEF Future of Jobs Report 2025, Malaysia profile, p. 169). A typical career tool today gives no time-based, sourced answer: **0** of the steps in a static skills list say when a skill becomes exposed to AI.
Core Measurable KPI	**Plan completion rate:** % of test users who go from CV upload to ranked future roles *and* an ordered, sourced reskilling plan in 5 minutes or less. **Next-step clarity:** % of test users who, after using the app, can name one specific skill to learn next and a free resource for it (asked before and after use).
Phase 2 Impact Target	In a user test with at least 20 XMUM students and staff by the final presentation: **≥ 80%** plan completion in 5 minutes or less, and an **increase of at least 40 percentage points** in next-step clarity (before vs after). Technical target: 100% of recommended courses link-checked within 7 days of the demo (currently 53 of 53, checked 24 Oct 2026).

Sources: Hays Asia Salary Guide 2025 and MOSTI figures as cited in the SDC 2026 Domain 1 problem statement; World Economic Forum, Future of Jobs Report 2025, Malaysia economy profile; UN SDG Global Indicator Framework (target 4.4 wording).

5. Readiness Eligibility Checklist

☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☑ All fields in this template have been completed within the word limits stated.

6. IP Declaration & Originality

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon? **No.** The repository was created on 21 October 2026, after the hacking period began, and all project code was written during the hackathon.

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public datasets, or pre-trained AI models that your team did not create? **Yes** – all declared and used within their licences:

Name	Type	License
FastAPI, Uvicorn, Pydantic, python-multipart, httpx	Code (backend)	MIT / BSD-3-Clause

pypdf	Code (PDF reading)	BSD-3-Clause
React, Vite, Tailwind CSS	Code (frontend)	MIT
scikit-learn, pandas, matplotlib (training only)	Code	BSD-3-Clause / PSF-based
GPTs-are-GPTs task exposure labels (Eloundou et al., 2023)	Dataset	MIT
O*NET task statements	Dataset	CC BY 4.0
arXiv API metadata (paper counts only)	Dataset	arXiv API Terms of Use (metadata CC0)
WEF Future of Jobs Report 2025; DOSM Salaries & Wages and Labour Force data; TalentCorp MyCOL 2024/2025	Dataset (published figures, cited in the app)	Public reports, cited with page numbers
Google Gemini API (optional CV reading)	Model (API)	Gemini API Terms of Service

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon? **No.**

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Azimkhan Bekassyl

Azimkhan Bekassyl, Team Leader, BEBZA

Date: 24 October 2026